MicroMod edge, reset, boot, test pads

File: sheets/edge.kicad_sch

RT1176 I/O

File: sheets/mcu.kicad_sch

RT1176 power

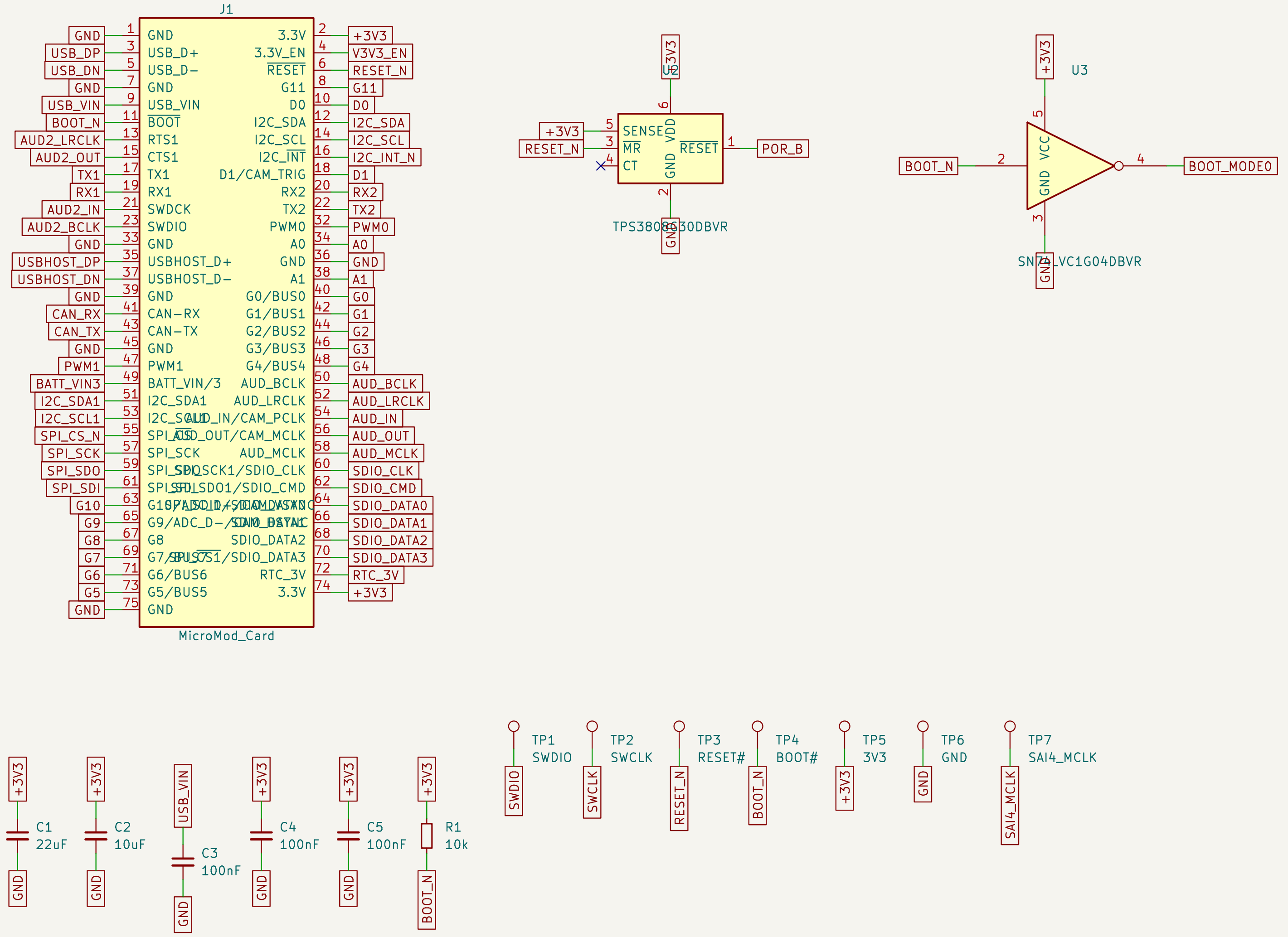
File: sheets/power.kicad_sch

Flash, clocks, boot straps, LED

File: sheets/mem.kicad_sch

MicroMod processor card, NXP i.MX RT1176
markrlupton
Sheet: /
File: rt1176-micromod-som.kicad_sch
Title: RT1176 MicroMod SOM
Size: A4 Date:
Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/5

MicroMod processor card edge: SparkFun MicroMod Teensy pinout (docs/PINOUT.md).
Pins 13/15/21/23 carry SAI4 (AUD2) instead of RTS1/CTS1/SWDCK/SWDIO; SWD is on test pads TP1/TP2.
RESET# -> U2 manual reset; U2 holds POR_B low through power-up (open drain, POR_B has an internal 100k pull-up).
BOOT# -> BOOT_MODE1 and, inverted by U3, BOOT_MODE0: released = 10 internal boot, held = 01 USB serial downloader.

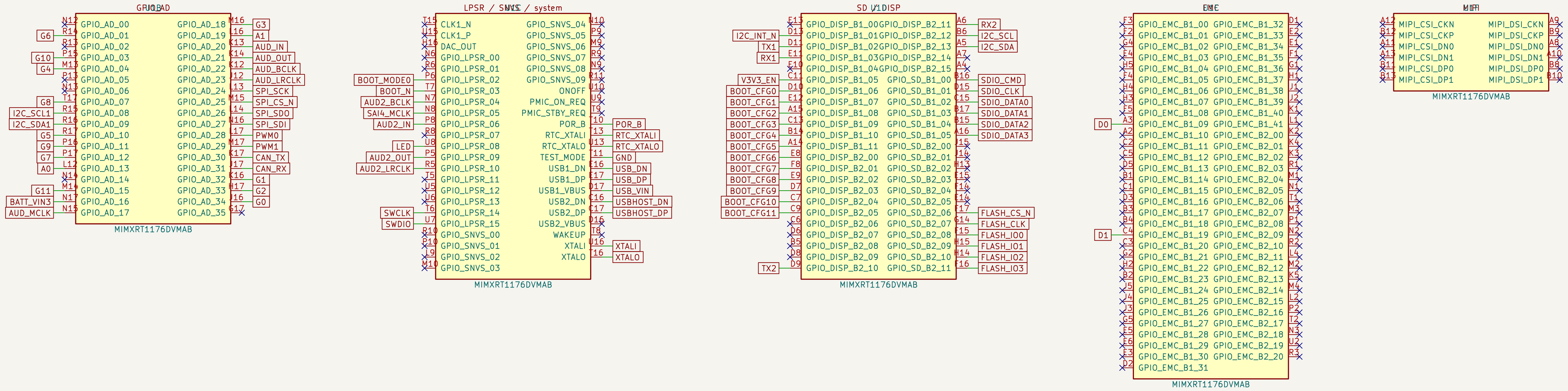


Generated by tools/schgen.py; do not edit by hand
markrlupton
Sheet: /MicroMod edge, reset, boot, test pads/
File: edge.kicad_sch

Title: RT1176 MicroMod SOM: MicroMod edge, reset, boot, test pads

Size: A3 Date: **Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 2/5

All I/O banks at 3.3 V. Unused pads float (pad keepers). BOOT_CFG pads only go to the straps on the Flash sheet.



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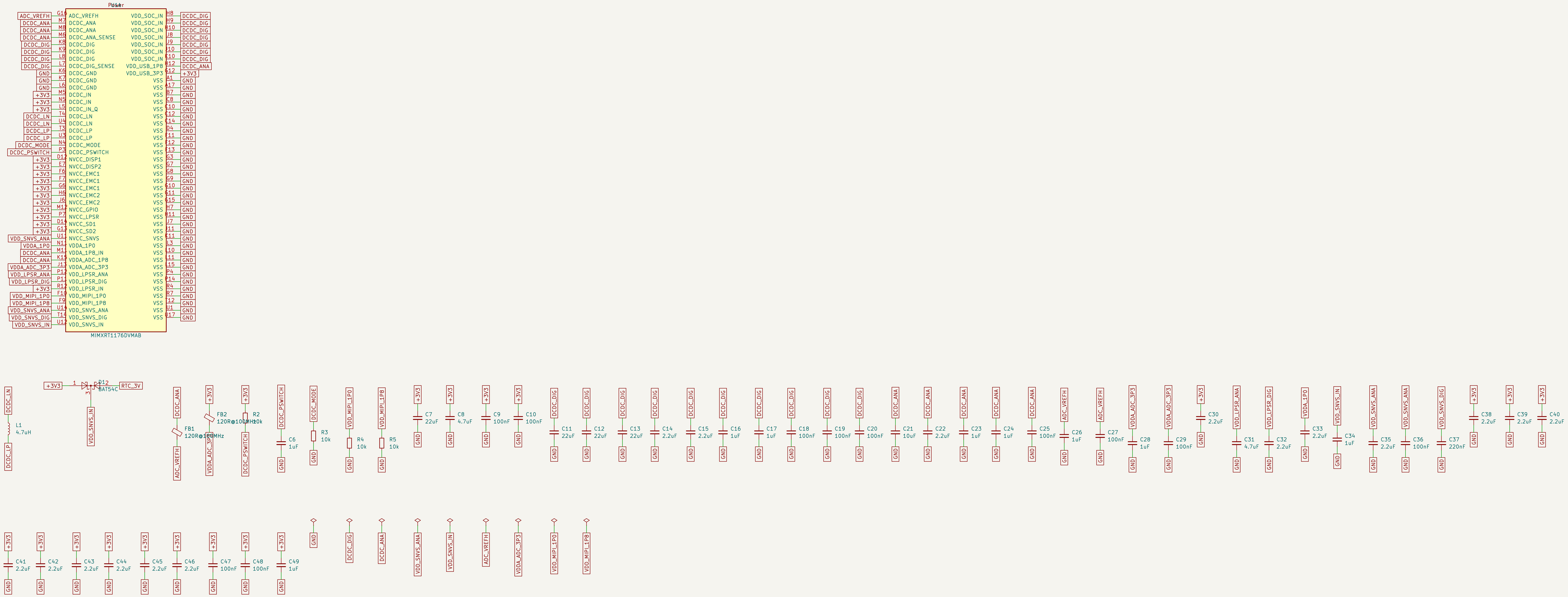
Sheet: /RT1176 I/O/
File: mcu.kicad_sch

Title: RT1176 MicroMod SOM: RT1176 I/O

Size: A2	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

3.3 V only from the carrier. RT1176 internal DCDC makes DCDC_DIG (1.0/1.1 V core) and DCDC_ANA (1.8 V).
0402 parts sit under the BGA at the ball named in their Note field; 0603/0805 bulk next to the BGA.



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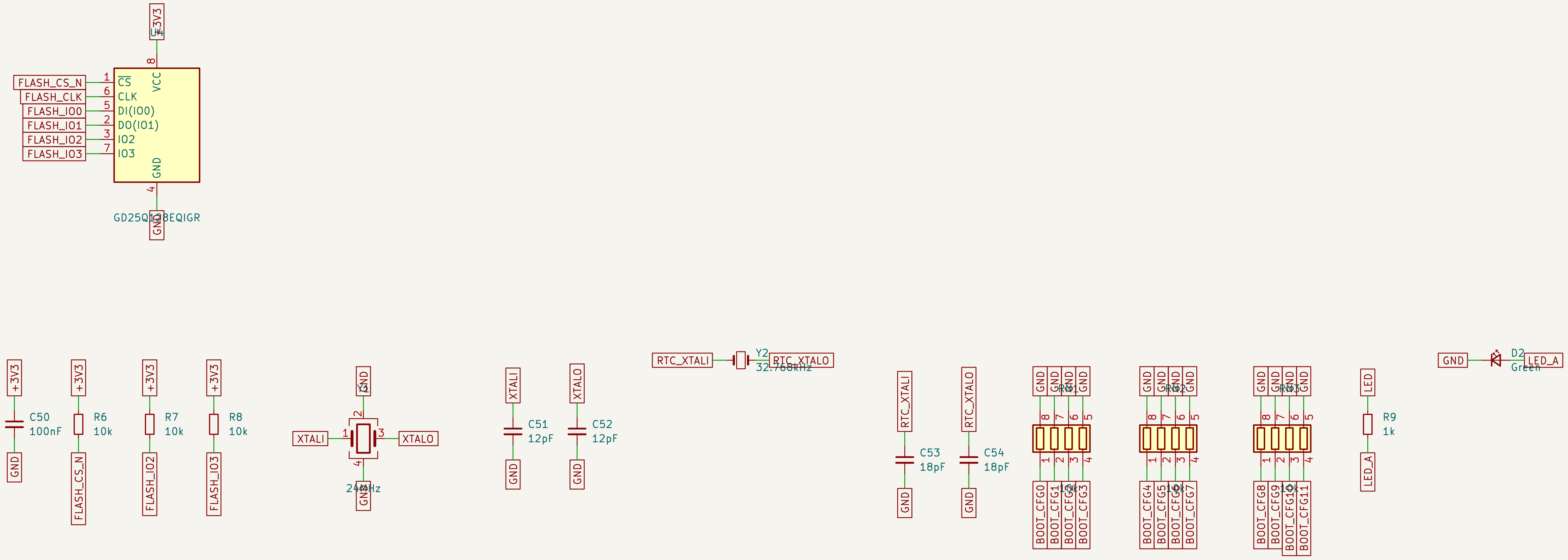
Sheet: /RT1176 power/
File: power.kicad_sch

Title: RT1176 MicroMod SOM: RT1176 power

Size: A2 Date: **Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/5

FlexSPI1 port A: the boot ROM default. DQS (GPIO_SD_B2_05) is left open for loopback read sampling.
BOOT_CFG straps all 0 = boot device FlexSPI NOR (used only while BT_FUSE_SEL = 0).



Generated by tools/schgen.py; do not edit by hand
markrlupton
Sheet: /Flash, clocks, boot straps, LED/
File: mem.kicad_sch

Title: *RT1176 MicroMod SOM: Flash, clocks, boot straps, LED*

Size: A3	Date:	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/5